

Recording a route without satellites: vibration-derived speed and its failure modes

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August 2026

Abstract

A phone can record where it has been without satellite positioning, by reading speed from how the vehicle shakes and direction from its motion sensors. We describe the method as shipped in a consumer iOS workout application, and report its accuracy on twenty-three instrumented journeys by car, motorcycle and on foot, each recorded with GPS running alongside purely as ground truth. On fast, steady driving with the phone coupled to the vehicle the recorded distance is within 3.8% over 15 km and heading within 5°; in slow stop-start traffic the same system reads 20–60% long. We show that this gap is not a defect but a property of the signal: absolute speed error is a few km/h at any speed, which is 1% of a motorway pace and 140% of a walking one. We give equal space to eight approaches that were implemented, measured and rejected, including three defeated by a single cause — Apple’s Core Motion attitude filter absorbs sustained linear acceleration as a change in the gravity direction, which erases exactly the signal that aircraft dead reckoning, acceleration-derived heading and inertial takeoff detection all depend on.

1 Introduction

Satellite positioning fails where routes are most often wanted: tunnels, underground car parks, dense urban canyons, and aircraft cabins. The standard response is inertial dead reckoning — integrate acceleration twice — which is well known to diverge, since accelerometer bias is indistinguishable from sustained real acceleration and its double integral grows without bound.

An alternative exists for wheeled vehicles. Road and engine vibration scale with speed, so a spectral signature of vertical acceleration can be mapped to speed directly, without integration and therefore without accumulation. CarSpeedNet [1] reports RMSE 1.8 m/s from 13.2 hours of accelerometer-only driving. That result is in-distribution: its test set is drawn from the same collection as its training data.

This paper describes such a system built into a shipping application, and — more usefully — reports what happens outside the distribution, on a phone that is

sometimes mounted, sometimes pocketed, sometimes in a hand, in a car, on a motorcycle, and once in an aircraft.

2 Method

2.1 Signature

Vertical acceleration $x[n]$ is sampled at 50 Hz into a ring buffer of $N = 256$ samples (4.0 s), windowed and transformed:

$$w[n] = \frac{1}{2} - \frac{1}{2} \cos\left(\frac{2\pi n}{N-1}\right) \quad (1)$$

$$X[k] = \sum_{n=0}^{N-1} (x[n] - \bar{x}) w[n] e^{-2\pi i k n / N} \quad (2)$$

Energy is accumulated into nine logarithmically spaced bands bounded at 0.4, 0.8, 1.6, 2.5, 4, 6, 9, 13, 18, 24 Hz:

$$f_b = \ln\left(\sum_{k \in b} |X[k]|^2 + \epsilon\right), \quad b = 1 \dots 9 \quad (3)$$

Two time-domain terms complete an eleven-dimensional signature:

$$f_{10} = \ln \sigma(x), \quad f_{11} = \ln\left(\frac{1}{N-1} \sum_n |x[n] - x[n-1]|\right) \quad (4)$$

The window length was measured rather than assumed. At 4 s the held-out bias is +1.0 km/h at 2–10 km/h; doubling to 8 s raises it to +6.0, because the window then straddles changes in speed. Finer low-frequency banding changed nothing.

2.2 Speed

No functional form is imposed. The system stores $(\mathbf{f}, v)$ pairs whenever GPS supplies a speed, and answers by locality. With running feature means μ_i and variances σ_i^2 ,

$$d^2(\mathbf{f}, \mathbf{g}) = \sum_{i=1}^{11} \left(\frac{f_i - \mu_i}{\sigma_i} - \frac{g_i - \mu_i}{\sigma_i} \right)^2 \quad (5)$$

$$\hat{v} = \frac{\sum_{j=1}^K w_j v_j}{\sum_{j=1}^K w_j}, \quad w_j = \frac{1}{d_j^2 + \epsilon} \quad (6)$$

with $K = 12$. Crucially the estimator is allowed to decline:

$$\min_j d_j^2 > 4 \implies \text{no answer} \quad (7)$$

Error scales directly with match distance — MAE 5.3 km/h below $d^2 = 1.03$, rising to 16.9 in the range 2.12–4.36 — so refusing a distant match removes the worst readings.

2.3 Regime separation

A store holding several vehicles holds several mutually incompatible mappings. Measured directly: a ground-trained model reported 19 km/h at a 900 km/h cruise, because cruise is *quieter* than taxiing and the slowest stored signature was the closest match. After flight data entered the same store, a city drive reported a mean of 337 km/h against a true 30. Air and ground are therefore stored and searched as separate partitions.

2.4 Direction

Turn angle comes from the gyroscope, absolute direction from the magnetometer:

$$\theta_t = \theta_{t-1} + \Delta\psi_{\text{gyro}} + \kappa(\psi_{\text{datum}} + \beta - \theta_{t-1}) \quad (8)$$

where β is the *carry offset*: the angle between where the phone points and where the body travels. β is not observable without an external reference, so it is learned from GPS course over the first 180 s of a workout and then frozen — which mirrors the physical case the mode exists for: signal on the way to the aircraft, none thereafter.

2.5 Projection

$$\Delta s = \hat{v} \Delta t, \quad \phi_t = \phi_{t-1} + \frac{\Delta s \cos \theta_t}{R}, \quad \lambda_t = \lambda_{t-1} + \frac{\Delta s \sin \theta_t}{R \cos \phi_t} \quad (9)$$

Only the first point of a route originates from GPS.

3 Results

3.1 Distance

Figure 1 gives the headline. A 15 km open-road drive records to +3.8%; a four-minute car-park trip to +61.6%. The ordering is by average speed, not by any property of the route.

3.2 Why slow journeys read long

Figure 2 resolves the apparent contradiction. On a store built from matched journeys, held-out bias is +1.0 to +1.7 km/h at *every* speed. A constant absolute error is a constant *relative* error only if speed is constant;

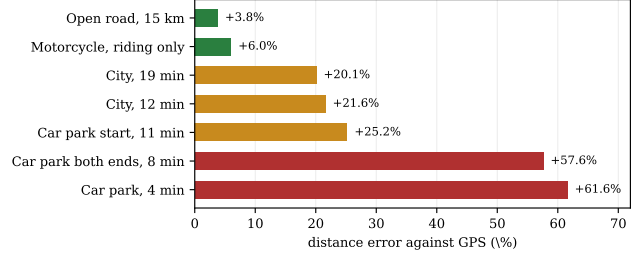


Figure 1: Recorded distance against GPS ground truth, seven journeys. The system is accurate when fast and steady, and reads long in slow traffic.

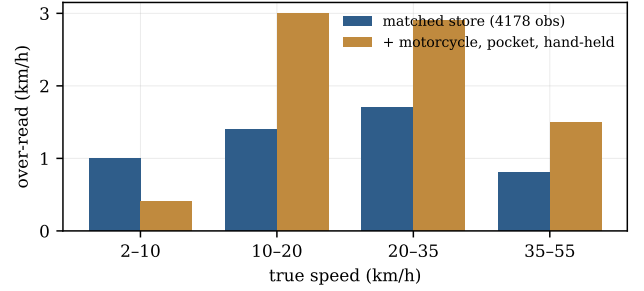


Figure 2: Leave-one-out bias against true speed. A matched store is accurate to 1–2 km/h across the range; adding other carry positions and vehicles roughly doubles it.

at 50 km/h it is 1%, at 5 km/h it is over 100%. The percentage figures in Figure 1 are therefore a statement about the journeys, not about the estimator.

This also explains why no correction helps. Subtracting a constant 6 km/h — fitted on two drives where the bias was +6.0 and +6.8 — repaired those two and degraded three others, reducing total absolute error by only 7%.

3.3 Direction

With β unlearned, two journeys showed mean signed errors of +24.9° and −35.8° with circular concentration $R = 0.93$ and 0.71: a near-constant rotation, not drift. The recorded shape and length were correct — one 15 km route matched its true net displacement to within 8 m of magnitude while pointing 26° wrong, finishing 4174 m from the true endpoint. Re-integrating with β applied moves the finish to 366 m, 2.4% of path length. After the bounded warm-up shipped, four consecutive journeys measured +0.8°, −1.3°, +1.0° and +2.1°.

4 Negative results

We report these in full because each cost real effort and three share one cause.

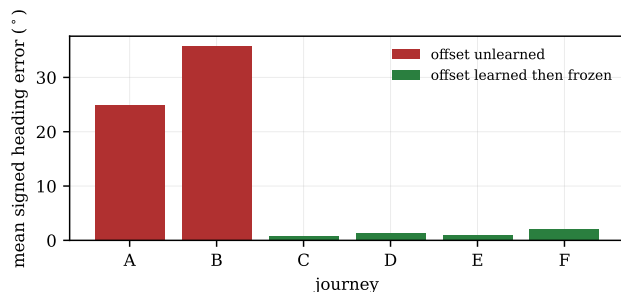


Figure 3: Mean signed heading error. Before β was learned, routes were correctly shaped but pivoted about their origin.

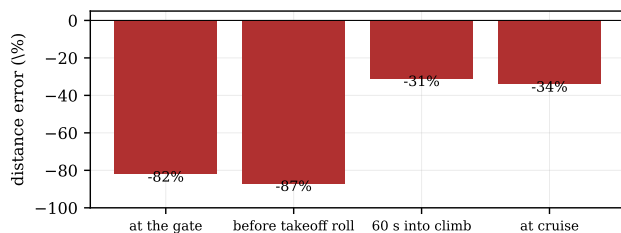


Figure 4: Replay of a real 22.8-minute flight with GPS removed at four points. Losing signal on the ground — the normal case — freezes taxi speed for the entire flight.

4.1 Aircraft speed cannot be integrated

A flight was recorded with GPS valid on all 1323 ticks to 706 km/h. Replaying it with GPS removed (Figure 4) gives -82% to -87% distance when signal is lost on the ground. The along-track integral recovers approximately 60 km/h of a 679 km/h climb: after 400 s of held speed the correction contributed -2 km/h.

4.2 One cause, three failures

Core Motion’s attitude filter reinterprets sustained linear acceleration as a change in the gravity direction. A takeoff roll is the most sustained acceleration a phone will ever experience, and is therefore precisely what the filter removes. This defeated:

1. *Aircraft speed integration*, above.
2. *Deriving β from acceleration*. A vehicle accelerates along its axis, so the world-frame acceleration bearing should give the travel direction. Measured against known answers on three drives, circular concentration was $R = 0.15, 0.18, 0.19$ — noise.
3. *Inertial takeoff detection*. Integrating positive along-track acceleration and latching at 45 m/s in one unbroken episode should identify a takeoff and nothing else. Replayed: today’s car 52.5 m/s (latched), a second car 209.9 (latched), a motorcycle 328.5 (latched), a two-minute *walk* 119.7 (latched), and

the actual flight 15.4 — never latched. Backwards on every case. The episode rarely breaks in a vibrating vehicle because road noise holds the signal above the floor, so the integral accumulates the positive half of the noise indefinitely.

4.3 Regime identification does not transfer

Whole-session fingerprints — the mean normalised signature of a workout — separate regimes cleanly. Across eight sessions, same-vehicle-and-carry pairs sit 0.54–1.70 apart, same-vehicle-different-carry 1.81–2.87, and different-vehicle 4.30–6.97.

Restricting the neighbour search by that fingerprint nevertheless made prediction *worse* ($+2.2 \rightarrow +3.4$ km/h). The ranking is unreliable at the individual level: for one car journey the nearest stored session was a car (1.38), but the second was a *hand-held* session (1.54) and the third a *motorcycle in a pocket* (1.74), both ahead of the same car driven the same morning (2.29).

4.4 Confidence signals that do not work

Neighbour disagreement is the natural confidence measure for a k -NN regressor. It fails here in the wrong direction: out-of-distribution motorcycle queries had a *tighter* neighbour spread than in-distribution car queries (5.2 vs 12.4 km/h), because they land consistently in one wrong region. Session-level fingerprint distance correlated with speed bias at $r = -0.01$ over six journeys.

4.5 Discriminating a hijacked pedometer

On a motorcycle, engine vibration is counted as footsteps. Apple’s activity classifier reported **automotive** on 636 ticks and **walking** on 271 of one continuous ride, and each flicker allowed the pedometer to take over: it drove 27% of the ride, reporting a mean 3.3 km/h while GPS recorded 49.6.

Three per-tick discriminators were measured against real walking and rejected: vibration energy (79% caught at 21% false), gait-band spectral concentration (distributions overlap), and both combined with rotation rate (69% at 7%). What worked was history rather than instantaneous features: across two real walks the classifier *never* reported automotive in 190 stepping ticks, while on motorcycles the median gap since it last did was 84 s. Requiring two minutes of silence before a pedometer-driven revocation suppressed 61% of false hand-overs at zero measured cost.

5 Systems failures

Three defects were not algorithmic, and each silently invalidated data for weeks.

A stale reading is not a missing reading. `CLLocationManager` suspends heading updates outside the foreground, and the cached property is never cleared. On an 81-minute drive the heading datum held *one value for 4790 of 4832 seconds* while the device physically rotated 5264° . A fallback to Core Motion’s heading had been added for the missing case and was never reached, because it was written as `clHeading ?? motionHeading` and a frozen value is not `nil`.

Background permission gating. `allowsBackgroundLocationUpdates` was enabled only under *Always* authorisation. Under *When In Use* the process was suspended mid-journey: a single 371-second hole in a 21-minute drive, during which the 50 Hz stream delivered 49 samples where 18,550 were due, accounting for the entire 20% distance shortfall.

Instrumentation that cannot express the failure. A 17-minute workout the user believed was recording without GPS had the mode active for 34 seconds; 63 GPS fixes entered the route. The guarantee held — the code path returns early — but nothing in the log stated the mode’s state, and it took an hour to establish from a side effect. Two columns now record it directly.

6 Limitations

- **Hand or pocket carry destroys the signal.** At matched speeds a hand-held phone reads 0.6–0.8 lower in the summary feature, and the feature stops varying with speed altogether: 7.10, 7.08, 7.07, 7.13, 7.25 across 10 to 65 km/h. Reported speed becomes a constant near 15–25 km/h regardless of truth.
- **Stationary vehicles with running engines.** A motorcycle at a red light reads 5.2 km/h, accumulating 690 m over 339 stationary ticks. No tested signal separates an idling engine from a slow crawl.
- **Car-park ramps.** A vehicle crawling a concrete spiral shakes as hard as one at 40 km/h on a road; measured error is +169% to +297%. Detection requires two signals jointly — barometric climb above 0.08 m/s *and* signed net rotation above 150° over 30 s — since the barometer alone flags every hill and rotation alone every junction.
- **Unseen vehicles.** The model answers only for conditions it has observed.

7 Conclusion

Vibration-derived speed is a sound basis for GPS-free route recording within its regime: fast, steady travel with the phone mechanically coupled to the vehicle, where we measure 3.8% distance error over 15 km and 5° of heading error. Outside that regime the failures are not gradual — a hand on the phone removes the signal entirely, and an aircraft’s defining acceleration is erased by the platform’s own sensor fusion.

We would emphasise two lessons over any result. First, the dominant error sources across this work were not modelling choices but sensor-platform semantics: a suspended process, a stale cached heading, an attitude filter making a defensible choice that happens to destroy the signal of interest. Second, an estimator that reports without expressing doubt is dangerous in a way that a wrong estimator is not; every failure above persisted for as long as it did because the system reported it with the same confidence as a correct reading. The shipped application now exposes its own error bars to the user.

References

- [1] CarSpeedNet: A deep neural network for vehicle speed estimation from smartphone accelerometer. arXiv:2401.07468.